

Johnny Dunn

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Summary

Full-stack engineer with deep expertise in production AI/ML systems, building and scaling ETL pipelines with RAG, and fine-tuning / optimizing LLMs. Experienced with automated evaluation and optimization of generative AI models and pair programming frameworks.

Professional Experience

Machine Learning Engineer

Independent Consulting - Contract | Remote | May 2019 – Current

- Developed multi-tenant CRM features and API integrations in Next, tRPC, Drizzle, Supabase for Tasty (4+ month contract).
- Built unified ML pipeline for Linkby (6-month project) automating feature engineering and model training from SQL for advertising data, enabling parallel training of RF, SVM, Naive Bayes, and LR models with A/B testing framework.
- Improved AI text detection accuracy from 55% to 71% for Grapple Research (5-month contract) by fine-tuning GPT-3.5 on expanded, deduplicated dataset. Reduced training costs by 64% with truncation & summarizing based on article structure.
- Developed document matching tool for ImmenseX (4-month engagement) reducing initial prompt writing by 60% for devs. Deployed RAG pipeline using Weaviate, Neo4j (hybrid vector and graph-based search) for conversational memories.
- Achieved 77% sentiment analysis accuracy (random forest) trained on 10K texts for HereAfterLegacy.ai; built out initial MVP to scrape scripts from TV shows to recreate digital persona of 3 fictional characters (14-month contract).
- Trained neural style transfer models for public exhibit applying film-inspired filters to selfies for Edelman (3-month project).
- Created user research, animated wireframes, and functional demo for MVP at Notebrush (15-month engagement).

Senior Blockchain Engineer

Tilting Point | New York, NY | Nov 2021 – Feb 2023

- Developed ERC-20/721 smart contracts with upgradability / proxy patterns, built and deployed backend REST APIs for NFT marketplace, and internal airdropping tool. Rescued \$1M MATIC through proxy contract methods regenerating Gnosis safes.
- Architected blockchain state gRPC microservices, supporting 4+ games with 1000+ events/sec (on/off-chain throughput).

Software Engineer 2

eBay Inc. | New York, NY | Jun 2020 – Apr 2021

- Built REST APIs in Scala / Spring-Boot showing new ads in real-time from clicking selected item options (in under 1s latency).
- Handled E2E ML delivery infrastructure (K8s + Jenkins) enabling teams to deploy models into sandboxed UIs; accelerated deep learning team iteration by 50% allowing ads team to validate changes without waiting for releases from other teams.

Cognitive Implementation Engineer

Amelia.ai | New York, NY | Jun 2017 – Dec 2017

- Implemented features like PII redaction for HIPAA, SOC 2 compliance for AI assistant serving Fortune 500 medical suppliers.

Projects

AgentOS – TypeScript runtime for orchestrating adaptive AI agents with emergent behaviors. github.com/framersai/agentos
tenets – Intelligently aggregates context for LLM prompts with multi-factor relevance and neural ranking. github.com/jddunn/tenets
MagicLogger – Color logger for Node / browser; universal style schema & OpenTelemetry support. github.com/manicinc/magiclogger
Restless - Hierarchical attention network model for anti-malware; 93% accuracy, [ClaMP](https://huggingface.co/datasets/cla-mp) dataset. github.com/jddunn/restless

Education

Bachelor of Fine Arts in Game Design (May 2017) – Parsons School of Design

Awards

5th Place, AT&T National VR and AR Challenge, with THiNKIMMERSIVE team. (<http://npe100.com/> / mobile app), 2017.

Skills

Programming Languages	Python, TypeScript, JavaScript, Go, Scala, SQL, Solidity
Frontend	React, Next.js, Redux, Vue, Nuxt, Tailwind, D3.js, three.js
Backend	FastAPI, Django, Node.js (Express, NestJS), gRPC, GraphQL, RabbitMQ, Kafka
AI / ML	PyTorch, TensorFlow, Keras, LangChain, LangGraph, LlamaIndex, Transformers, Hugging Face, Scikit-learn, spaCy, Pandas, NumPy, OpenCV, CUDA, XGBoost, Stable Diffusion, OpenAI, DSPy
Databases / DevOps	PostgreSQL, SQLAlchemy, Supabase, Redis, Neo4j, Pinecone, MongoDB, Elasticsearch, AWS (Lambda, EC2, EBS, RedShift), Docker, Kubernetes, Jenkins, GitHub Actions
Tools / Platforms / Other	Unity3D, Adobe Creative Suite, Electron, Capacitor, Android Studio, Quasar, Selenium, Puppeteer